

# Power Consumption Analytics Dashboard

Project Documentation

## End-to-End Data Analytics Project

### Technology Stack

SQL Server | SQL Server Management Studio (SSMS) | Microsoft Excel | Power BI | DAX

Prepared as portfolio documentation for the completed analytics project.

## 1. Project Overview

The **Power Consumption Analytics Dashboard** is an end-to-end data analytics and business intelligence project developed to analyze electricity consumption, customer activity, billing, revenue, and geographic patterns. The workflow combines SQL Server for structured data management and analysis, Microsoft Excel for supporting analysis, and Power BI for interactive visualization and customer-level exploration.

## 2. Project Objectives

- Build and manage a structured relational database for electricity consumption and billing data.
- Use SQL for data preparation, validation, aggregation, and analytical querying.
- Use Excel Pivot Tables and charts for supporting analysis and validation.
- Build an interactive Power BI report for monitoring revenue, consumption, customers, and billing-related indicators.
- Support analysis from high-level KPIs down to individual customer details.

## 3. Dataset and Database

The project database is **PowerConsumptionDB**. It contains structured data for customers, customer categories, meters, meter readings, tariffs, bills, states, and cities.

| Metric              | Value                                                 |
|---------------------|-------------------------------------------------------|
| Customers           | 100                                                   |
| Meters              | 100                                                   |
| Meter Readings      | 1,200                                                 |
| Bills               | 1,200                                                 |
| States              | 8                                                     |
| Cities              | 18                                                    |
| Customer Categories | 4 - Residential, Commercial, Industrial, Agricultural |

## 4. SQL and Data Processing

SQL Server and SSMS were used to create and manage the database, populate and validate records, and perform analytical queries. The repository separates database setup scripts, analytical queries, views, and the database backup to keep the SQL workflow reproducible and easy to review.

A dedicated view, **vw\_DashboardData**, provides prepared analytical data for downstream reporting. The SQL component demonstrates database creation, relational data handling, validation, aggregation, analytical querying, views, and stored-procedure work.

## 5. Excel Analysis

The Excel workbook **Power\_Consumption\_Analytics.xlsx** contains three principal sheets: **Dashboard**, **Pivot Tables**, and **Raw\_Data**. Excel is used as a supporting analysis and validation layer through KPI summaries, Pivot Tables, Pivot Charts, and filtering.

## 6. Power BI Report

The Power BI file **Power Consumption Project PowerBI.pbix** contains three report pages. The report combines KPI cards, charts, slicers, tables, DAX measures, and drill-through functionality to move from summary analysis to customer-level detail.

- **Page 1:** Overall dashboard with core revenue, consumption, customer, geographic, and trend indicators.
- **Customer Analysis:** Customer-focused analysis for comparing customer activity, revenue, and consumption.
- **Customer Details:** Drill-through page designed for detailed analysis of a selected customer.

## 7. Key Dashboard Metrics

| KPI                    | Value               |
|------------------------|---------------------|
| Total Revenue          | INR 2.79M (approx.) |
| Total Units Consumed   | 333.99K             |
| Total Customers        | 100                 |
| Average Units Consumed | 278.33              |

## 8. Key Analytical Insights

- The analytical model covers 100 customers and 100 meters, enabling both aggregate and customer-level analysis.
- The dashboard summarizes approximately INR 2.79 million in revenue and 333.99K units of recorded electricity consumption.
- Revenue can be compared geographically across states and across different customer categories.
- Payment-status analysis provides visibility into billing and payment distribution.
- Monthly revenue visualization makes changes and patterns over time easier to identify.
- Customer analysis and drill-through functionality connect high-level dashboard KPIs with detailed individual-customer information.

## 9. Skills Demonstrated

- SQL and relational database management
- Data validation, aggregation, and analytical querying
- SQL views and stored procedures
- Excel Pivot Tables, Pivot Charts, KPI analysis, and filtering
- Power BI data modeling and dashboard development
- DAX measures and KPI calculations
- Interactive slicers, filtering, and drill-through
- Data visualization and business intelligence reporting

## **10. Repository Deliverables**

The repository is organized into database scripts and backup, data folders where applicable, the Excel analysis workbook, the Power BI report, project documentation, and selected screenshots. This structure presents the complete workflow from database preparation to final dashboard reporting.

## **11. Conclusion**

This project demonstrates a practical end-to-end analytics workflow using SQL Server, Excel, and Power BI. It shows the ability to organize structured data, perform analytical processing, validate results, develop interactive dashboards, and communicate consumption, customer, billing, and revenue information through business-oriented visual reporting.